

Derivative Basics: Guide

The big idea. The derivative $f'(a)$ is the **slope of the tangent line** at $x = a$. It is the **instantaneous rate of change**: what the average rate $\frac{\Delta y}{\Delta x}$ approaches as the two points slide together. The definition is a limit; the shortcut rules come from it.

USING THE LIMIT DEFINITION

- 1 Write $f(x + h)$.** Replace every x in the formula with $(x + h)$. Use parentheses.

2 Build the quotient. $\frac{f(x + h) - f(x)}{h}$. Subtract the *whole* $f(x)$.

3 Expand and combine. Everything without an h should cancel, leaving h on every term.
- 4 Factor out and cancel h .** The h in the denominator must go before you take the limit.

5 Let $h \rightarrow 0$. Any term still holding an h disappears.

6 Check. Test the answer against a power-rule result.

Most common mistakes: (1) Writing $f(x + h) = f(x) + h$. You must substitute $x + h$ into the formula. (2) Forgetting that $(x + h)^2 = x^2 + 2xh + h^2$. (3) Letting $h \rightarrow 0$ **before** cancelling the h (gives $\frac{0}{0}$). (4) Using $f(a)$ in place of the slope: the **value** $f(a)$ and the **slope** $f'(a)$ are different numbers.

WHAT IS THE QUESTION ASKING?

THE QUESTION SAYS	IT MEANS	DO THIS
Slope of the tangent at $x = a$	$f'(a)$	Differentiate, then plug in a.
Equation of the tangent line	Point and slope	$y - f(a) = f'(a)(x - a)$.
Where is the tangent horizontal?	$f'(x) = 0$	Solve $f'(x) = 0$.
Instantaneous rate of change	Derivative	Not an average. Use f' at that moment.
Maximum or minimum	Critical points	Solve $f'(x) = 0$, then check the sign of f'.

FORMULA BANK

LIMIT DEFINITION General $f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$ At a point $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$	BASIC RULES Constant $\frac{d}{dx} c = 0$ Power $\frac{d}{dx} x^n = nx^{n-1}$ Const. multiple $\frac{d}{dx} cf = cf'$ Sum $\frac{d}{dx} (f \pm g) = f' \pm g'$ Rewrite roots and fractions as powers first: $\sqrt{x} = x^{1/2}, \frac{1}{x^2} = x^{-2}$.	TANGENT LINE Equation $y - f(a) = f'(a)(x - a)$ Slope from f' . The point $(a, f(a))$ comes from the original function.
GRAPH OF F VS F' $f' > 0$: f is increasing . $f' < 0$: f is decreasing . $f' = 0$: horizontal tangent (possible max or min). f' changes $+$ $\rightarrow$ $-$: local max . $- \rightarrow +$: local min .	NOT DIFFERENTIABLE WHEN... A corner (sharp point, like $ x $). A cusp or vertical tangent . A discontinuity . Differentiable $\Rightarrow$ continuous. The reverse is false .	OPTIMIZATION PLAN 1. Write the quantity to maximize or minimize. 2. Use the constraint to get one variable . 3. Set the derivative to 0 and solve. 4. Test it, and answer what was asked.

Eight problems that cover the exam

Think slope: the derivative is the slope of the tangent line.

Walkthrough 1: Limit definition

EXPAND, CANCEL H

Use the definition to find $f'(x)$ for $f(x) = x^2 + 3x$.

1 F(X+H) $(x+h)^2 + 3(x+h) = x^2 + 2xh + h^2 + 3x + 3h$

2 SUBTRACT $f(x+h) - f(x) = 2xh + h^2 + 3h$

3 DIVIDE $\frac{h(2x + h + 3)}{h} = 2x + h + 3$

4 LET $h \rightarrow 0$ $f'(x) = \boxed{2x + 3}$

Walkthrough 2: Tangent line

POINT AND SLOPE

Find the tangent line to $f(x) = x^3 - 2x$ at $x = 2$.

1 SLOPE $f'(x) = 3x^2 - 2$, so $f'(2) = 12 - 2 = 10$.

2 POINT $f(2) = 8 - 4 = 4$, so the point is $(2, 4)$.

3 WRITE IT $y - 4 = 10(x - 2)$, so $\boxed{y = 10x - 16}$

Walkthrough 3: Basic rules

REWRITE AS POWERS

Differentiate $f(x) = 4x^3 - 6\sqrt{x} + \frac{5}{x^2}$.

1 REWRITE $f(x) = 4x^3 - 6x^{1/2} + 5x^{-2}$

2 POWER RULE Multiply by the exponent, subtract 1 from it.

3 RESULT $f'(x) = 12x^2 - 3x^{-1/2} - 10x^{-3}$

4 TIDY $12x^2 - \frac{3}{\sqrt{x}} - \frac{10}{x^3}$

Walkthrough 7: Average vs instantaneous TWO DIFFERENT RATES

A position is $s(t) = t^2 + 2t$ feet. Find the average velocity on $[1, 3]$ and the velocity at $t = 1$.

1 AVERAGE $\frac{s(3) - s(1)}{3 - 1} = \frac{15 - 3}{2} = 6$ ft/s.

2 DERIVATIVE $s'(t) = 2t + 2$

3 AT $T=1$ $s'(1) = \boxed{4}$ ft/s. Not the same as the average.

Walkthrough 8: Optimization

ONE VARIABLE

Fence a rectangle with 100 ft of fence. Find the dimensions of maximum area.

1 SETUP $2x + 2y = 100$, so $y = 50 - x$.

2 AREA $A = xy = x(50 - x) = 50x - x^2$

3 DERIVATIVE $A' = 50 - 2x = 0$, so $x = 25$, $y = 25$.

4 CHECK A' goes $+$ $\rightarrow$ $-$ at 25: a max. Area = 625 ft^2 .

Walkthrough 4: Reading f and f'

SLOPE TELLS THE STORY

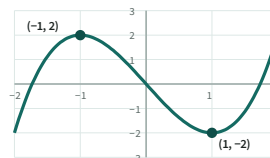
The graph shows $f(x) = x^3 - 3x$. Find where $f' = 0$ and the sign of f' .

1 FLAT SPOTS The tangent is horizontal at the peak and valley:
 $x = -1$ and $x = 1$.

2 POSITIVE f rises for $x < -1$ and $x > 1$, so $f' > 0$ there.

3 NEGATIVE f falls on $(-1, 1)$, so $f' < 0$ there.

4 CHECK $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$ agrees.



Where the tangent is flat

Walkthrough 5: Where is it not differentiable?

CORNERS

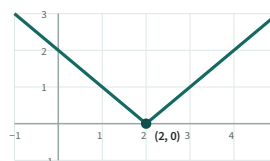
Graph shown: $f(x) = |x - 2|$. Is f differentiable at $x = 2$?

1 LEFT Left of 2 the slope is -1 .

2 RIGHT Right of 2 the slope is $+1$.

3 COMPARE The slopes disagree, so $f'(2)$ does not exist: a **corner**.

4 STILL... f is **continuous** at 2. Continuous does not mean differentiable.



A sharp point has no tangent line

Walkthrough 6: Critical points

SIGN CHART

Find and classify the critical points of $f(x) = x^3 - 3x^2$.

1 DERIVATIVE $f'(x) = 3x^2 - 6x = 3x(x - 2)$

2 SOLVE $f' = 0$ at $x = 0$ and $x = 2$.

3 SIGNS $f' > 0$ before 0, $f' < 0$ on $(0, 2)$, $f' > 0$ after 2.

4 CLASSIFY $+$ $\rightarrow$ $-$ at 0: local **max**, $f(0) = 0$. $-$ $\rightarrow$ $+$ at 2: local **min**, $f(2) = -4$.